

one of them.

- Q.30 What are histograms? Give their application.
- Q.31 Write a short note on sequential sampling plan.
- Q.32 What are the main characteristics of ISO 9000?
- Q.33 What are transducers? Give their classification.
- Q.34 Explain the measurement of pressure with the help of transducers.
- Q.35 What are the main advantages and application of LVDT?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain $\bar{X}$ and R chart in detail.
- Q.37 Explain normal, poisson and binomial distributions in detail.
- Q.38 What is TQM? How it is implemented? Explain TQM model and give the principles, benefit and list various techniques of TQM.

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6th Sem / Mechanical Engg. (MSIL) Sub : Inspection & Quality Control

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the followings is a control chart for variables?
 - a) $\bar{X}$ -Chart
 - b) U-Chart
 - c) C-Chart
 - d) All of the above
- Q.2 What does QA and QC stand for?
 - a) Quality Assurance & Queuing control
 - b) Quality Adjustment and Quality completion
 - c) Quality Assurance and Quality control
 - d) Quality Adjustment and Queuing control
- Q.3 The least count of micrometer is :
 - a) 0.001mm
 - b) 0.02mm
 - c) 0.002mm
 - d) 0.01mm
- Q.4 When the measurement system is not linear _____ is needed.
 - a) Process to be changed by which the units are made
 - b) Calibration of the measurement system
 - c) Change in the process parameters
 - d) No change

- Q.5 Variance is the _____ of standard deviation.
 a) Square root b) Cube root
 c) Square d) None of the above
- Q.6 Which of these is called the center of mass of the sample data?
 a) Sample variance
 b) Median of the data
 c) Arithmetic mean of the data
 d) The number having the highest frequency
- Q.7 ISO was founded in the year of _____
 a) 1943 b) 1946
 c) 1949 d) 1956
- Q.8 The full form of TQM is
 a) Total Quality Management
 b) Total Quantity Management
 c) Total Quotient Management
 d) Total Quality Mark
- Q.9 Self generating transducers are _____ transducers.
 a) Passive b) Active
 c) Inverse d) Secondary
- Q.10 A device which converts the energy from one form into another is known as _____.
 a) Sensor b) Capacitor
 c) Relay d) Transducer

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What are basic methods of taking measurements?
 Q.12 Define measurement.

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- Q.13 What is comparators?
 Q.14 Name any two types of errors.
 Q.15 Define median.
 Q.16 Give any two example of control charts for variables.
 Q.17 What is quality audit?
 Q.18 What is 5S?
 Q.19 What is passive transducer?
 Q.20 What is full form of LVDT?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define interchangeability. Explain its types.
 Q.22 What are the different standards for measurement? Explain.
 Q.23 Write a short note on pilot-piece inspection.
 Q.24 What is the use of callipers? Name the various types of calipers.
 Q.25 Write a short note on wire gauge.
 Q.26 Why are electrical measuring instruments more advantageous as compared to mechanical instrument?
 Q.27 Explain the process of testing and maintenance of measuring instruments.
 Q.28 Explain pneumatic comparators.
 Q.29 What are different sampling methods? Explain any

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